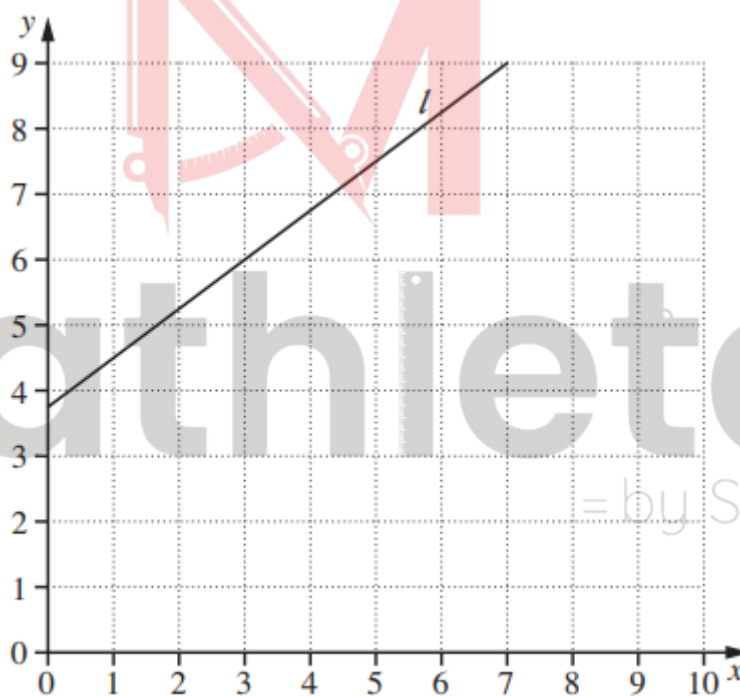


Worksheet 3 (Calculator)

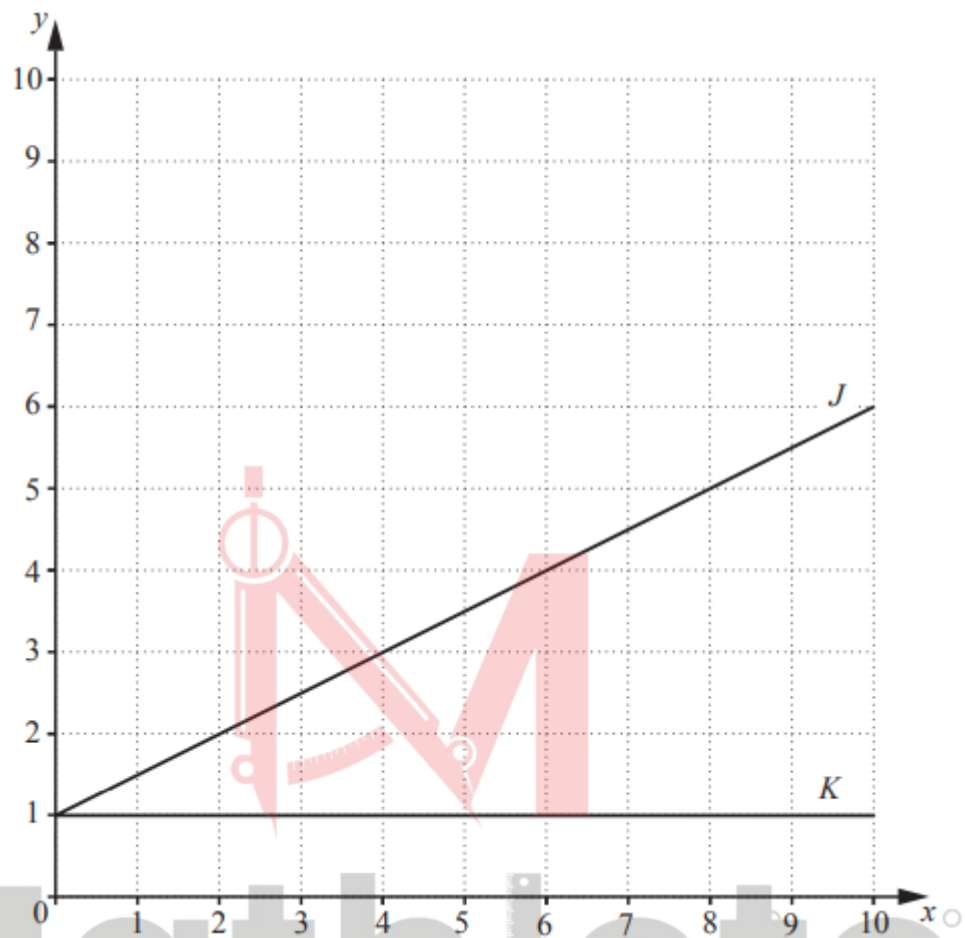
1. S11/qp21/q2

- a. A is the point (3, 6) and B is the point (11, 12). Find the coordinates of the midpoint of AB. [1]
- b. C and D have coordinates (10, 15) and (−8, −21).
i. Find the equation of the line CD in the form $y = mx + c$. [2]
ii. Does the point (−2, −9) lie on the line CD? Show your working to justify your answer. [1]
- c. The line l has equation $4y = 3x + 15$.
i.
a. Find the coordinates of the point where l crosses the x axis. [1]
b. Find the coordinates of the point where l intersects the line $y = p$. Express each coordinate in terms of p . [2]
ii. The line l is drawn on the grid below.



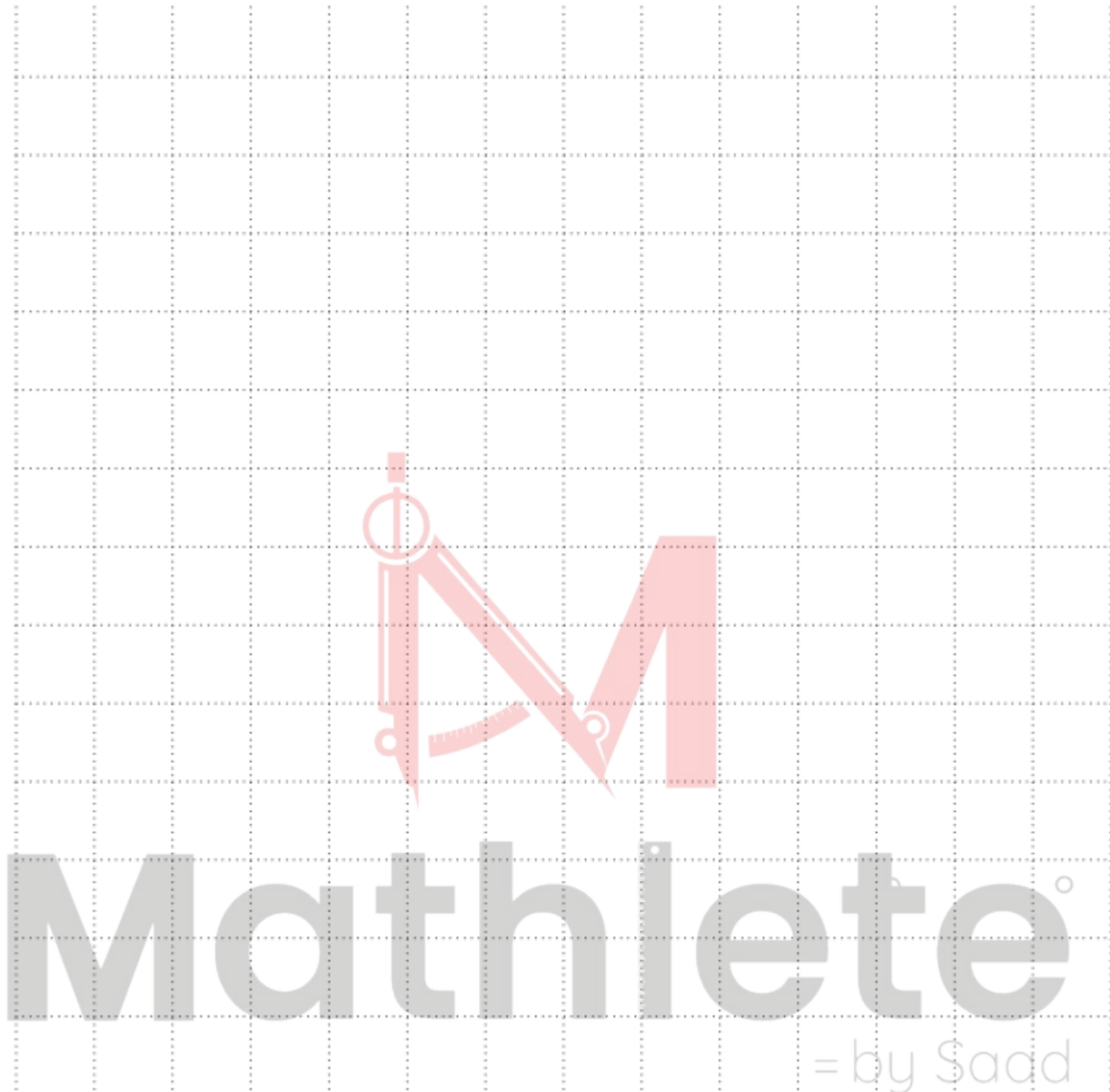
By drawing the line $3x + 2y = 30$ on the grid, find the coordinates of the point where these two lines intersect. [2]

2. S15/qp22/q1(b)



- i. Find the gradient of line J. [1]
- ii. Write down the equation of line K. [1]
- iii. Draw a line, L, through (6, 1) such that the area enclosed between J, K and L is 6 cm^2 . [1]
- iv. Find the equation of line L. [2]
- v. The line N is perpendicular to line J at (2, 2). Find the coordinates of the point where line N crosses the y-axis. [2]

3. W15/qp22/q2



A is the point (8, 7), B is the point (−2, 11) and C is the point (1, 7).

- a. Calculate the area of triangle ABC. [2]
- b. Calculate the length of AB. [2]
- c. Calculate the perimeter of triangle ABC [2]
- d. Calculate $\angle BAC$. [2]

4. S16/qp42/q9 (0580)

A line joins the points A $(-2, -5)$ and B $(4, 13)$.

- a. Calculate the length AB. [3]
- b. Find the equation of the line through A and B. Give your answer in the form $y = mx + c$. [3]
- c. Another line is parallel to AB and passes through the point $(0, -5)$. Write down the equation of this line. [2]
- d. Find the equation of the perpendicular bisector of AB. [5]

5. S17/qp41/q7 (0580)

A line joins the points A $(-3, 8)$ and B $(2, -2)$.

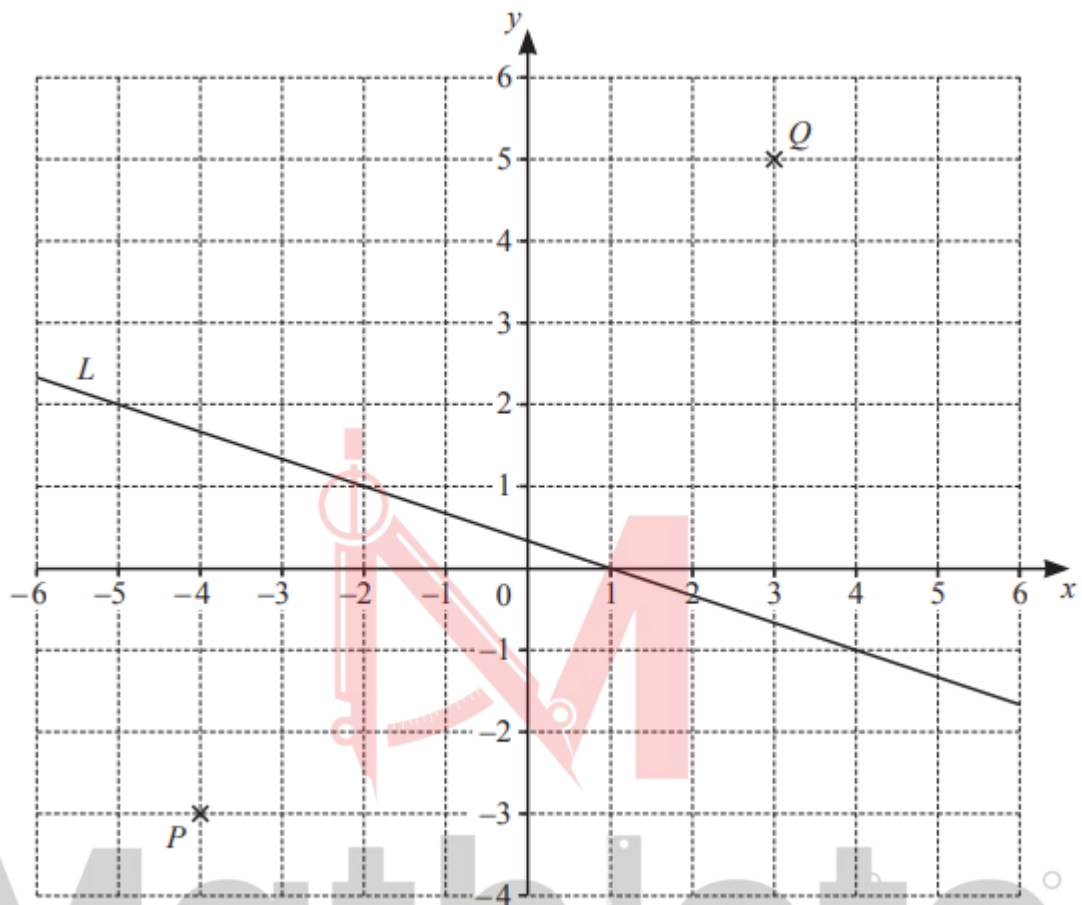
- a. Find the co-ordinates of the midpoint of AB. [2]
- b. Find the equation of the line through A and B. Give your answer in the form $y = mx + c$. [3]
- c. Another line is parallel to AB and passes through the point $(0, 7)$. Write down the equation of this line. [2]
- d. Find the equation of the line perpendicular to AB which passes through the point $(1, 5)$. Give your answer in the form $ax + by + c = 0$ where a, b and c are integers. [4]

6. W17/qp43/q8 (0580)

Line A has equation $y = 5x - 4$. Line B has equation $3x + 2y = 18$.

- a. Find the gradient of
 - i. line A, [1]
 - ii. line B. [1]
- b. Write down the co-ordinates of the point where line A crosses the x-axis. [2]
- c. Find the equation of the line perpendicular to line A which passes through the point $(10, 9)$. Give your answer in the form $y = mx + c$. [4]
- d. Work out the co-ordinates of the point of intersection of line A and line B. [3]
- e. Work out the area enclosed by line A, line B and the y-axis. [3]

7. W19/qp21/q10

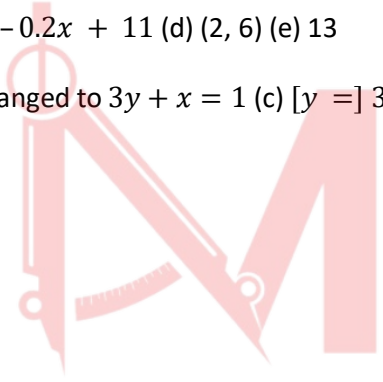


Points P and Q and the line L are shown on the grid.

- Calculate the length of PQ . [2]
- Show that the equation of line L is $3y + x = 1$. [3]
- Line M is perpendicular to line L and passes through point P . Find the equation of line M .
Give your answer in the form $y = mx + c$. [3]

Answer Key:

1. (a) (7, 9) (b) (i) $y = 2x - 5$ (ii) Yes and $-9 = 2 \times -2 - 5$ (c) (i) (a) $(-5, 0)$ (b) $(\frac{4p-15}{3}, p)$ (ii) $(5, 7\frac{1}{2})$
2. (i) 1 or 0.5 (ii) $y = 1$ (iii) Line from (6, 1) to (4, 3) (iv) $y = -x + 7$ (v) (0, 6)
3. (a) 14 (b) 10.8 (c) 22.8 (d) 21.8
4. (a) 19[.0] or 18.97 (b) $[y =] 3x + 1$ (c) $y = 3x - 5$ (d) $y = -\frac{1}{3}x + \frac{13}{3}$
5. (a) $(-0.5, 3)$ (b) $[y =] -2x + 2$ (c) $y = -2x + 7$ (d) $x - 2y + 9 = 0$ or $2y - x - 9 = 0$
6. (a) (i) 5 (a) (ii) $-\frac{3}{2}$ (b) $(\frac{4}{5}, 0)$ (c) $y = -0.2x + 11$ (d) (2, 6) (e) 13
7. (a) 10.6 (b) Gradient = $-1/3$ rearranged to $3y + x = 1$ (c) $[y =] 3x + 9$



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